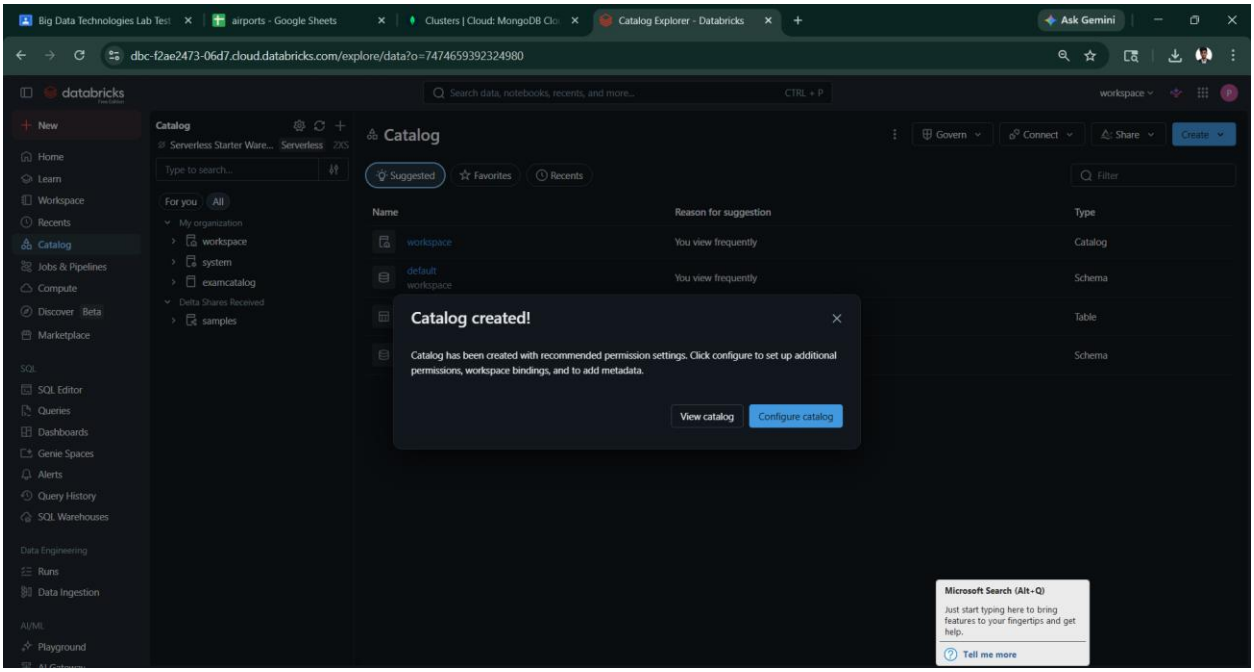
Ques 3

Spark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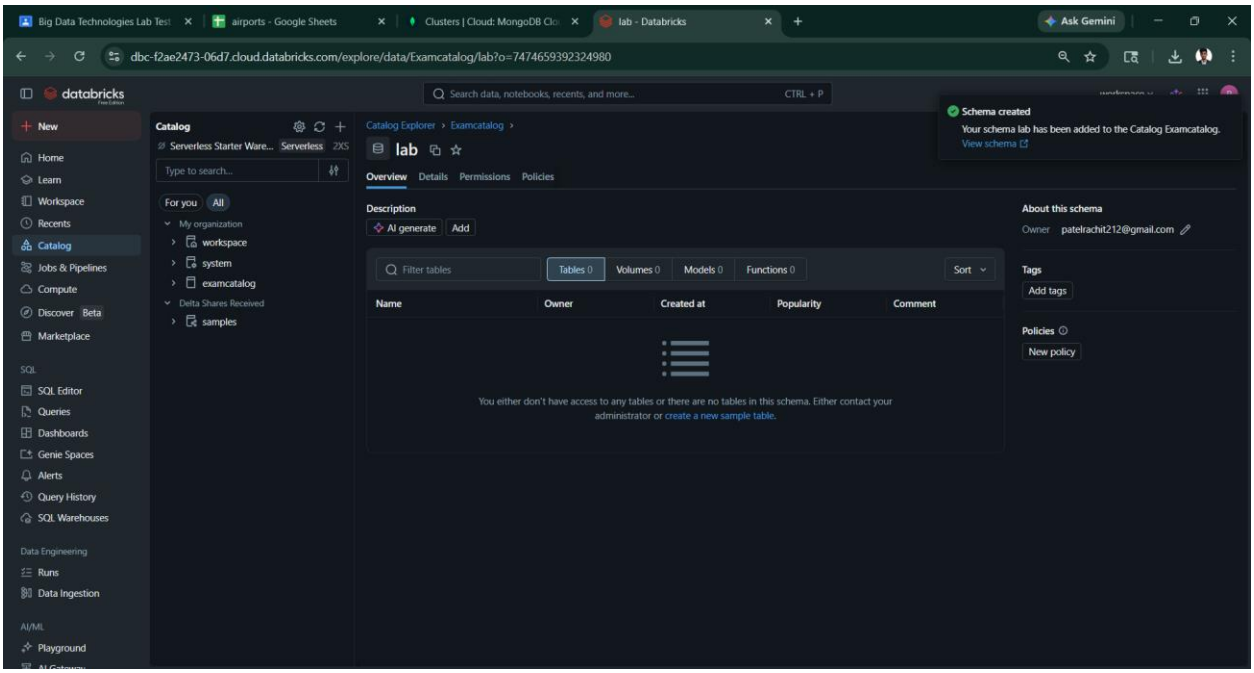
You view frequently
Schema
Table
Schema
Catalog created!
Catalog has been created with recommended permission settings. Click configure to set up additional permissions, workspace bindings, and to add metadata.
View catalog
Configure catalog
Microsoft Search (Alt+Q)
Just start typing here to bring features to your fingertips and get help.
Tell me more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Overview
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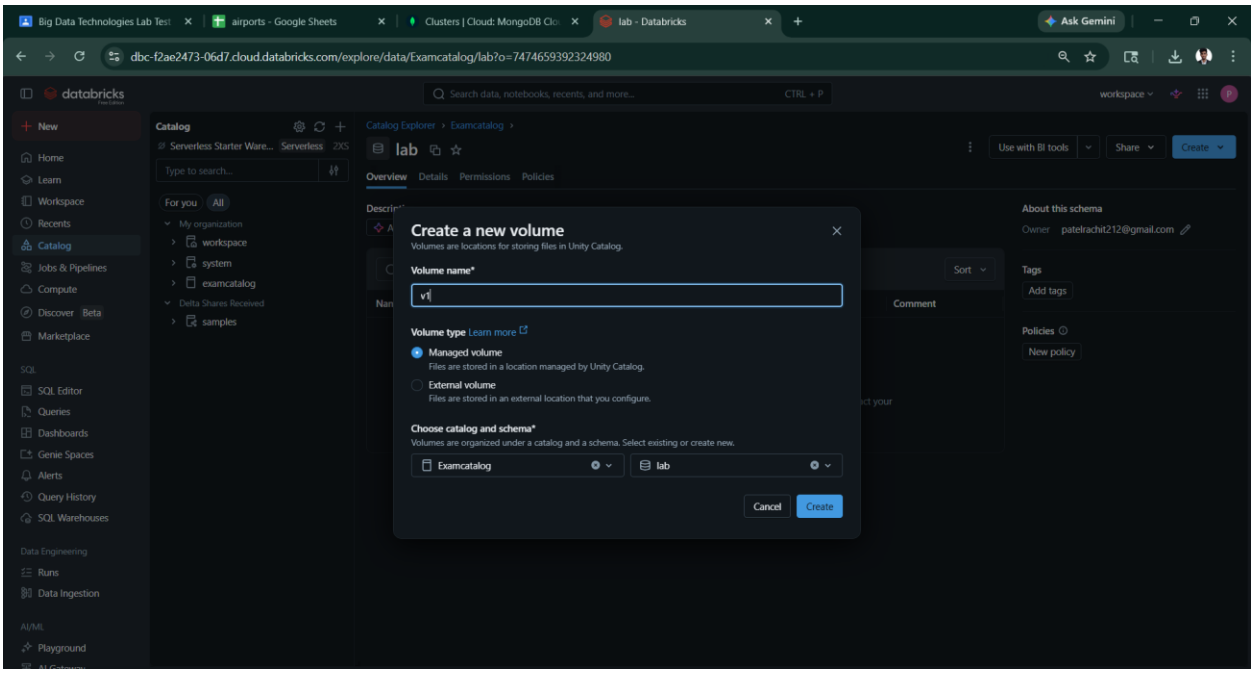
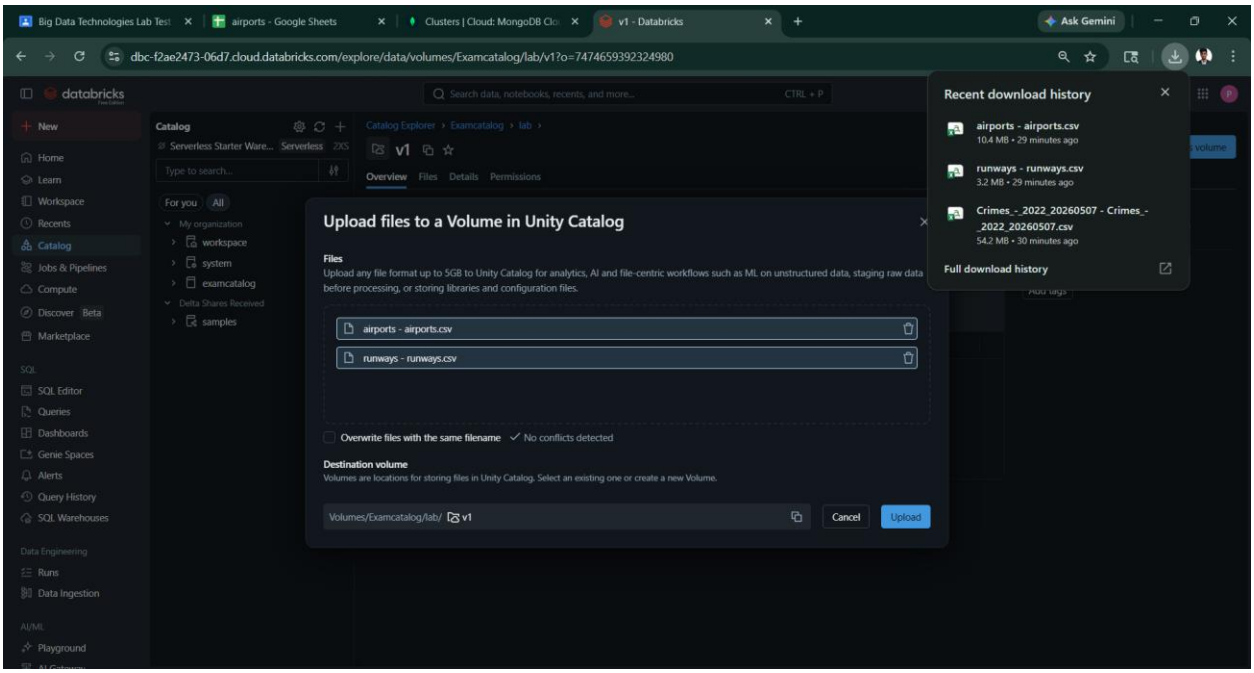
Policies
Schema created
Your schema lab has been added to the Catalog Examcatalog.
View schema
Description
AI generate
Add
Filter tables
Tables 0
Volumes 0
Models 0
Functions 0
Sort
Name
Owner
Created at
Popularity
Comment
You either don't have access to any tables or there are no tables in this schema. Either contact your administrator or create a new sample table.
About this schema
Owner patelrachit212@gmail.com
Tags
Add tags
Policies
New policy

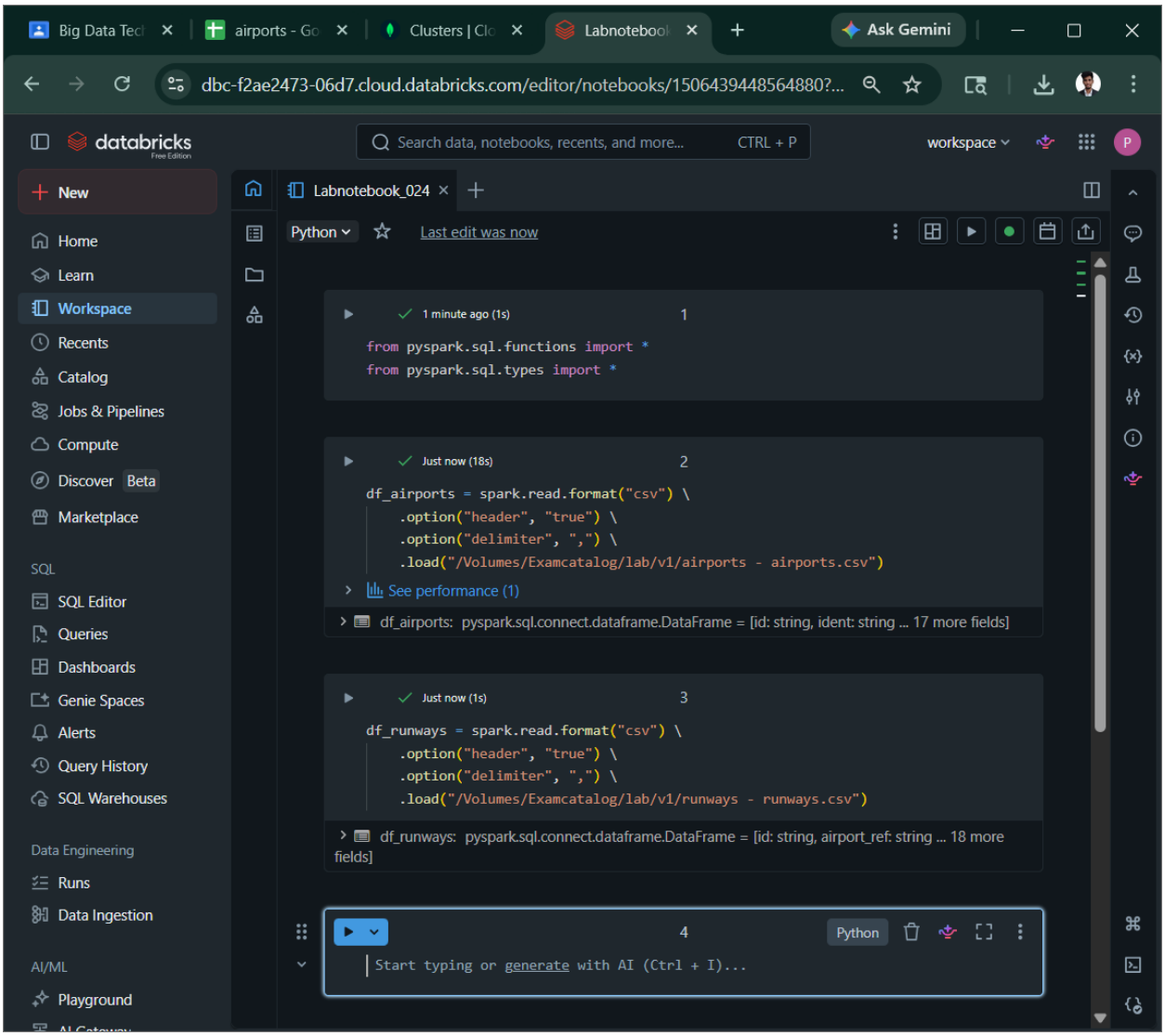


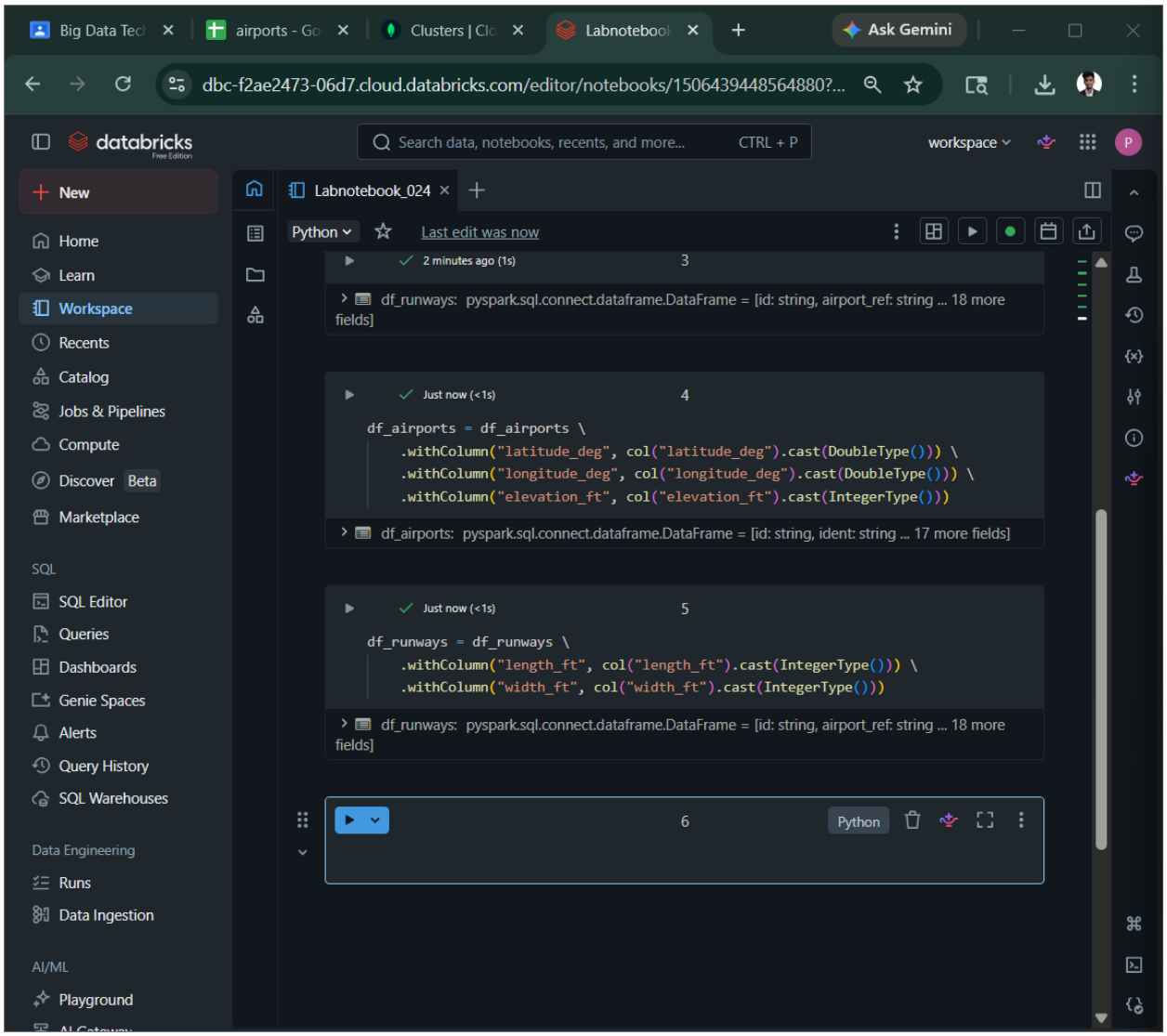

I have uploaded two csv in volume v1 then

Part A:

h. I loading the files



i. To cast columns




j. print the scehma and row count of both the dataframesz

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Python | Last edit was now

Just now (<1s) 6

```python
df_airports.printSchema()
```

```
root
 |-- id: string (nullable = true)
 |-- ident: string (nullable = true)
 |-- type: string (nullable = true)
 |-- name: string (nullable = true)
 |-- latitude_deg: double (nullable = true)
 |-- longitude_deg: double (nullable = true)
 |-- elevation_ft: integer (nullable = true)
 |-- continent: string (nullable = true)
 |-- iso_country: string (nullable = true)
 |-- iso_region: string (nullable = true)
 |-- municipality: string (nullable = true)
 |-- scheduled_service: string (nullable = true)
 |-- icao_code: string (nullable = true)
 |-- iata_code: string (nullable = true)
 |-- gps_code: string (nullable = true)
 |-- local_code: string (nullable = true)
 |-- home_link: string (nullable = true)
 |-- wikipedia_link: string (nullable = true)
 |-- keywords: string (nullable = true)
```

Just now (1s) 7

```python
print(df_airports.count())
```

```
85324
```

8 Python

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Python | Last edit was now

Just now (<1s) 8

```python
df_runways.printSchema()
```

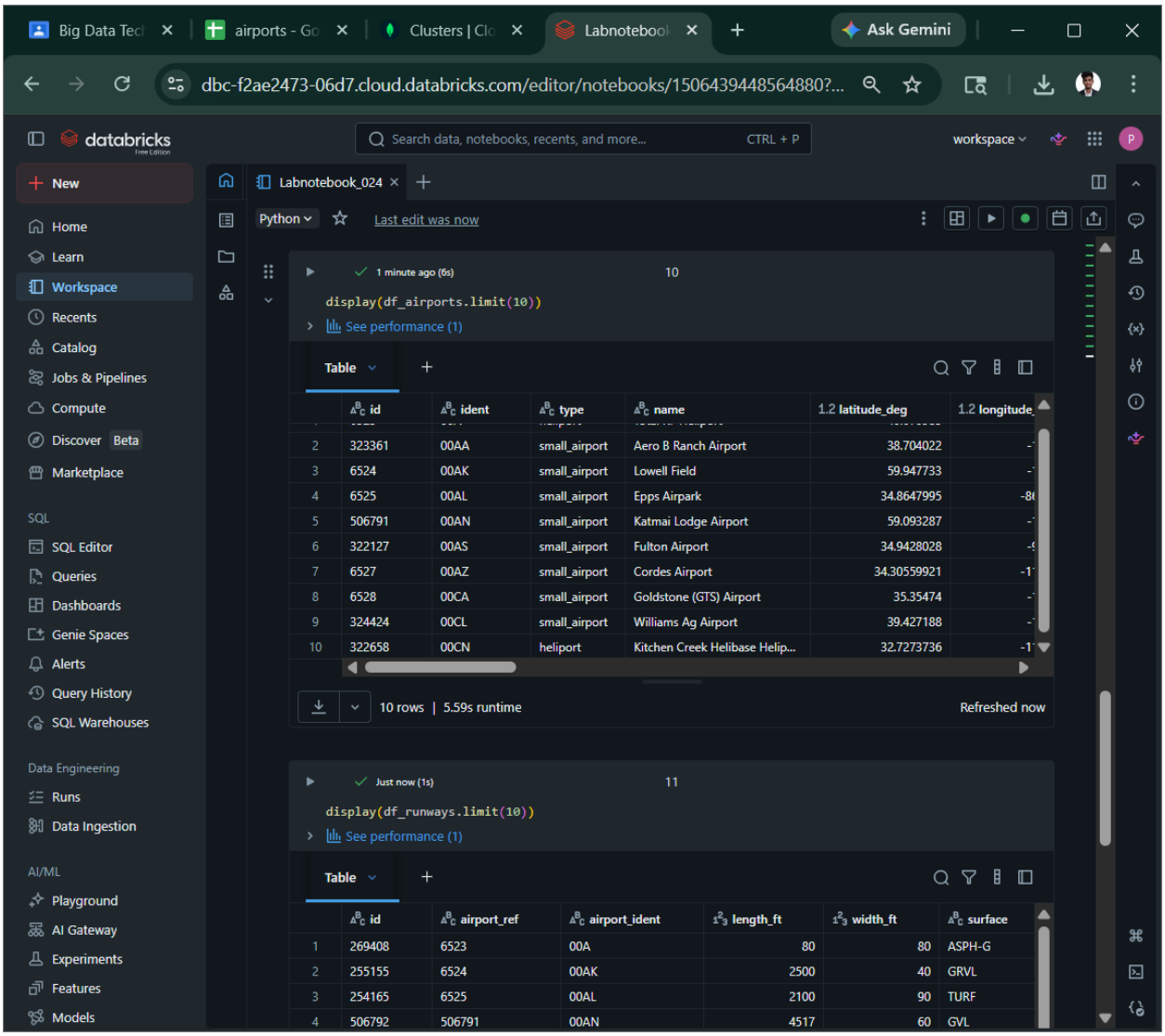
```
|-- id: string (nullable = true)
|-- airport_ref: string (nullable = true)
|-- airport_ident: string (nullable = true)
|-- length_ft: integer (nullable = true)
|-- width_ft: integer (nullable = true)
|-- surface: string (nullable = true)
|-- lighted: string (nullable = true)
|-- closed: string (nullable = true)
|-- le_ident: string (nullable = true)
|-- le_latitude_deg: string (nullable = true)
|-- le_longitude_deg: string (nullable = true)
|-- le_elevation_ft: string (nullable = true)
|-- le_heading_degT: string (nullable = true)
|-- le_displaced_threshold_ft: string (nullable = true)
|-- he_ident: string (nullable = true)
|-- he_latitude_deg: string (nullable = true)
|-- he_longitude_deg: string (nullable = true)
|-- he_elevation_ft: string (nullable = true)
|-- he_heading_degT: string (nullable = true)
|-- he_displaced_threshold_ft: string (nullable = true)
```

Just now (1s) 9

```python
print(df_runways.count())
```

```
47884
```

10 Python




part b

k. join df_airports with df_runways of the airport identifier colum (ident in df_airports matches airport_ident in df_runways ) using an inner join . name the result df_combined

l . add a new colum runway_category based on length_ft :"Long" if >=10000ft. "Medium" if between 5000 and 9999 ft and "short" if below 5000 ft

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Python | Last edit was now

1 minute ago (<1s) 12

```python
df_combined = df_airports.join(
    df_runways,
    df_airports.ident == df_runways.airport_ident,
    "inner"
)
```

df_combined: pyspark.sql.connect.dataframe.DataFrame = [id: string, ident: string ... 37 more fields]

Just now (<1s) 13

```python
df_combined = df_combined.withColumn(
    "runway_category",
    when(col("length_ft") >= 10000, "Long")
    .when((col("length_ft") >= 5000) & (col("length_ft") <= 9999), "Medium")
    .otherwise("Short")
)
```

df_combined: pyspark.sql.connect.dataframe.DataFrame = [id: string, ident: string ... 38 more fields]

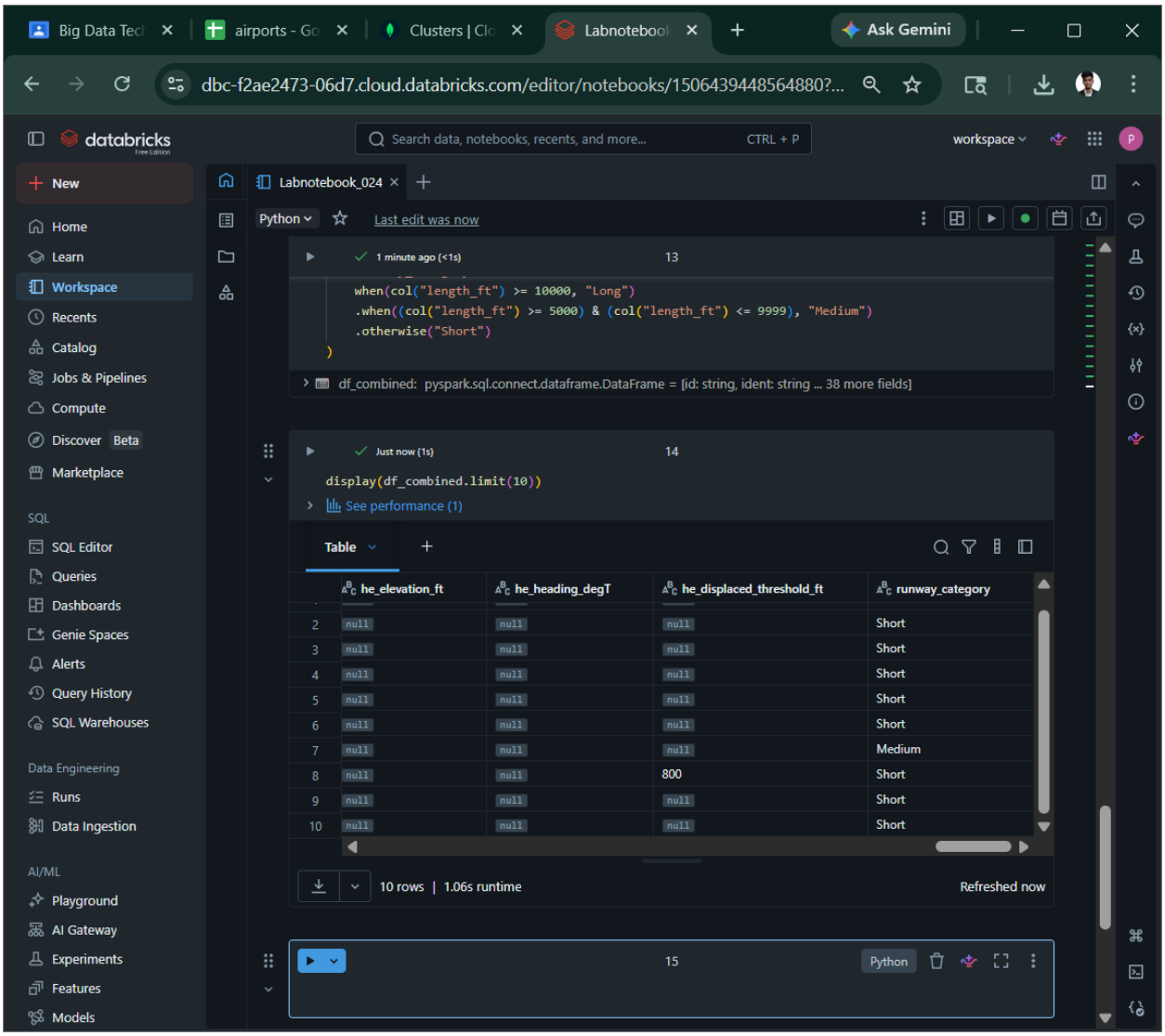
Just now (1s) 14

```python
display(df_combined.limit(10))
```

See performance (1)

Table

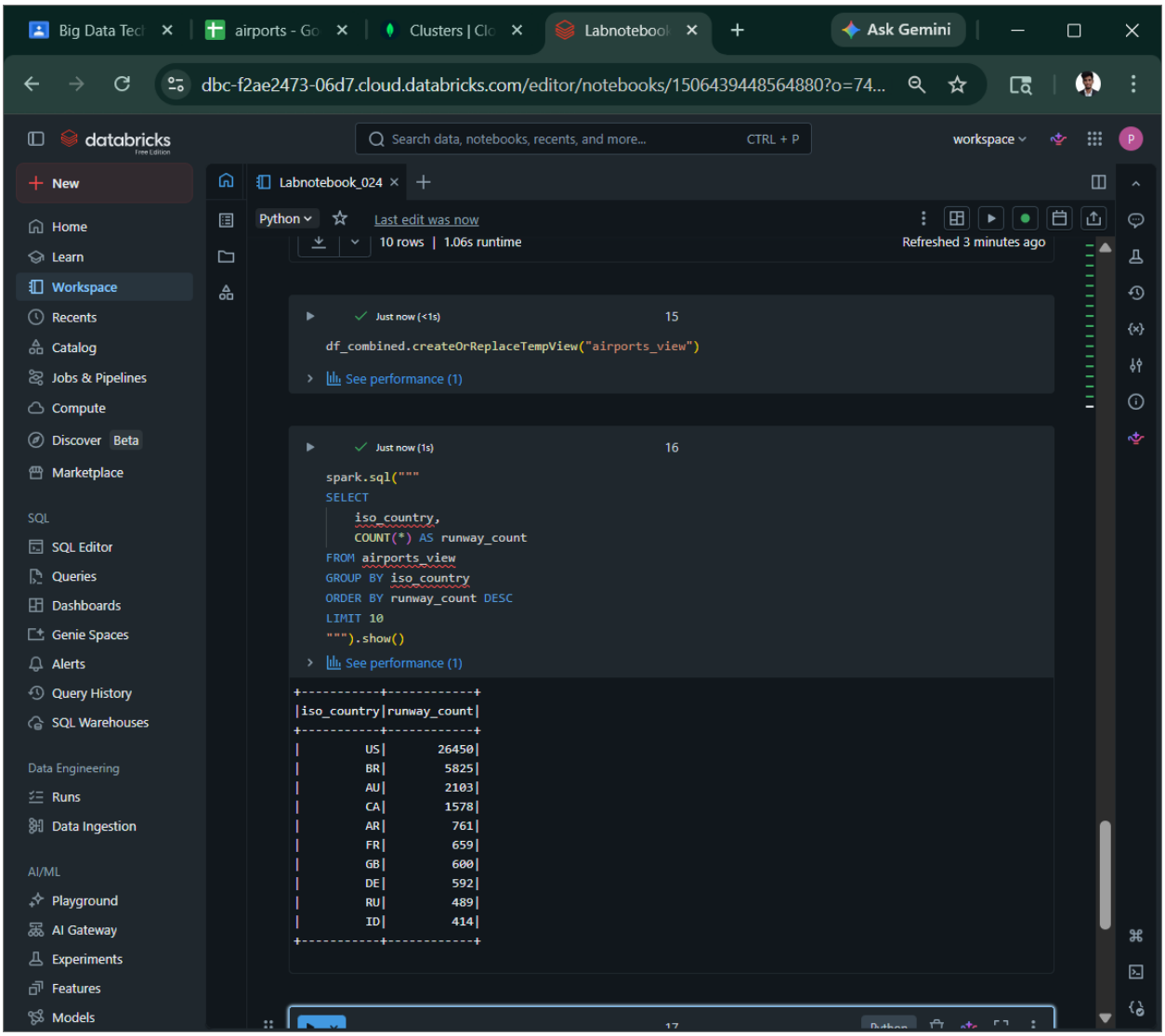
| | id | ident | type | name | latitude_deg | longitude_deg |
|---|---|---|---|---|---|---|
| 1 | 6523 | 00A | heliport | Total RF Heliport | 40.070985 | -74.933 |
| 2 | 6524 | 00AK | small_airport | Lowell Field | 59.947733 | -151.692 |
| 3 | 6525 | 00AL | small_airport | Epps Airpark | 34.8647995 | -86.77030 |
| 4 | 506791 | 00AN | small_airport | Katmai Lodge Airport | 59.093287 | -156.456 |
| 5 | 322127 | 00AS | small_airport | Fulton Airport | 34.9428028 | -97.8180 |
| 6 | 6527 | 00AZ | small_airport | Cordes Airport | 34.30559921 | -112.1650 |
| 7 | 6528 | 00CA | small_airport | Goldstone (GTS) Airport | 35.35474 | -116.885 |




part c analysis

create a temporary view named airports_view from df_combined and and write the following sql queries.

14. find the top 10 countries (iso_country) with the highest number of runways .diplsay country code and runway count,ordered descending




15. Find the average runway length (length_ft) per contient .Exclude rows where length_ft is null.Order by average length descending

Just now (1s) 17

```
spark.sql("""
SELECT
    continent,
    AVG(length_ft) AS avg_runway_length
FROM airports_view
WHERE length_ft IS NOT NULL
GROUP BY continent
ORDER BY avg_runway_length DESC
""").show()
```

See performance (1)

```
+---------+------------------+
|continent| avg_runway_length|
+---------+------------------+
|       AN| 7768.857142857143|
|       AS|6925.6469996647675|
|       AF| 6787.526448362721|
|       EU| 4103.151335311572|
|       OC|3558.0401785714284|
|       SA|2857.0891141239963|
|       NA| 2599.724218941758|
+---------+------------------+
```

16. Find all airports in india (iso_country-“IN”) that have at least one lighted runway (lighted=”1”)Display the arport name , municipality  and number of lighted runways.

Just now (1s) 18 Python

```python
spark.sql("""
SELECT
    name,
    municipality,
    COUNT(*) AS lighted_runways
FROM airports_view
WHERE iso_country = 'IN'
AND lighted = '1'
GROUP BY name, municipality
HAVING COUNT(*) >= 1
ORDER BY lighted_runways DESC
""").show(truncate=False)
```

See performance (1)

| name | municipality | lighted_runways |
|---|---|---|
| Indira Gandhi International Airport | New Delhi | 3 |
| Navi Mumbai International Airport | Navi Mumbai | 2 |
| Chennai International Airport | Chennai | 2 |
| Prayagraj Airport | Allahabad | 2 |
| Bidar Airport / Bidar Air Force Station | Bidar | 2 |
| Mangaluru International Airport | Mangaluru | 2 |
| Rajiv Gandhi International Airport | Hyderabad | 2 |
| Agra Airport / Agra Air Force Station | Agra | 2 |
| Tambaram Air Force Station | Chennai | 2 |
| Netaji Subhash Chandra Bose International Airport | Kolkata | 2 |
| Chhatrapati Shivaji Maharaj International Airport | Mumbai | 2 |
| Raja Bhoj International Airport | Bhopal | 2 |
| Kempegowda International Airport Bengaluru | Bengaluru | 2 |
| Ambala Air Force Station | Ambala | 2 |
| Dhulia Airport | NULL | 1 |
| Jammu Airport | Jammu | 1 |
| Raigarh Airport (JSPL) | NULL | 1 |
| Maharshi Valmiki International Airport | Faizabad | 1 |

17. list then top 5 airports (by name) with the longest maximum runway ength . display airport name, coutry and max runway length

Just now (1s) 19 Python

```python
spark.sql("""
SELECT
    name,
    iso_country,
    MAX(length_ft) AS max_runway_length
FROM airports_view
GROUP BY name, iso_country
ORDER BY max_runway_length DESC
LIMIT 5
""").show()
```

See performance (1)

```
+--------------------+-----------+-----------------+
|                name|iso_country|max_runway_length|
+--------------------+-----------+-----------------+
|Gunflint Seaplane...|         US|            30000|
|Libby Camps Seapl...|         US|            26000|
|Brookville Reserv...|         US|            25000|
|Long Lake Seaplan...|         US|            25000|
|Conchas Lake Seap...|         US|            21120|
+--------------------+-----------+-----------------+
```

/Volumes/examcatalog/lab/v1

Part d

Save as paqrquet

m. save the df_combined dataframe (with the runway_category colum) as a parquet file to your databrices volume.use overwrite mode

Last execution failed 20 Python

```python
df_combined.write.mode("overwrite").parquet(
    "/Volumes/Examcatalog/lab/v1/combined_airports_runways_parquet"
)
```

See performance (1) 1

```
[COLUMN_ALREADY_EXISTS] The column `id` already exists. Choose another name or rename the existing column. SQLSTATE: 42711
```

Diagnose error Debug Genie Code Quick Fix: ON

since same id is in both columns
rejoined df and selected columns

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Python | Last edit was now

Diagnose error | Debug | Genie Code Quick Fix: ON

Just now (<1s) 21

```python
df_combined = df_airports.alias("a").join(
    df_runways.alias("r"),
    col("a.ident") == col("r.airport_ident"),
    "inner"
).select(
    col("a.id").alias("airport_id"),
    col("a.ident"),
    col("a.type"),
    col("a.name"),
    col("a.latitude_deg"),
    col("a.longitude_deg"),
    col("a.elevation_ft"),
    col("a.continent"),
    col("a.iso_country"),
    col("a.iso_region"),
    col("a.municipality"),

    col("r.id").alias("runway_id"),
    col("r.airport_ref"),
    col("r.airport_ident"),
    col("r.length_ft"),
    col("r.width_ft"),
    col("r.surface"),
    col("r.lighted")
)
```

df_combined: pyspark.sql.connect.dataframe.DataFrame = [airport_id: string, ident: string ... 16 more fields]

Just now (3s) 22

```python
df_combined.write.mode("overwrite").parquet(
    "/Volumes/examcatalog/lab/v1/combined_airports_runways_parquet"
)
```

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Python — Last edit was now

Just now (3s) 22

```python
df_combined.write.mode("overwrite").parquet(
    "/Volumes/examcatalog/lab/v1/combined_airports_runways_parquet"
)
```

See performance (1)

Just now (1s) 23

```python
df_parquet = spark.read.parquet(
    "/Volumes/examcatalog/lab/v1/combined_airports_runways_parquet"
)
```

df_parquet: pyspark.sql.connect.dataframe.DataFrame = [airport_id: string, ident: string ... 16 more fields]

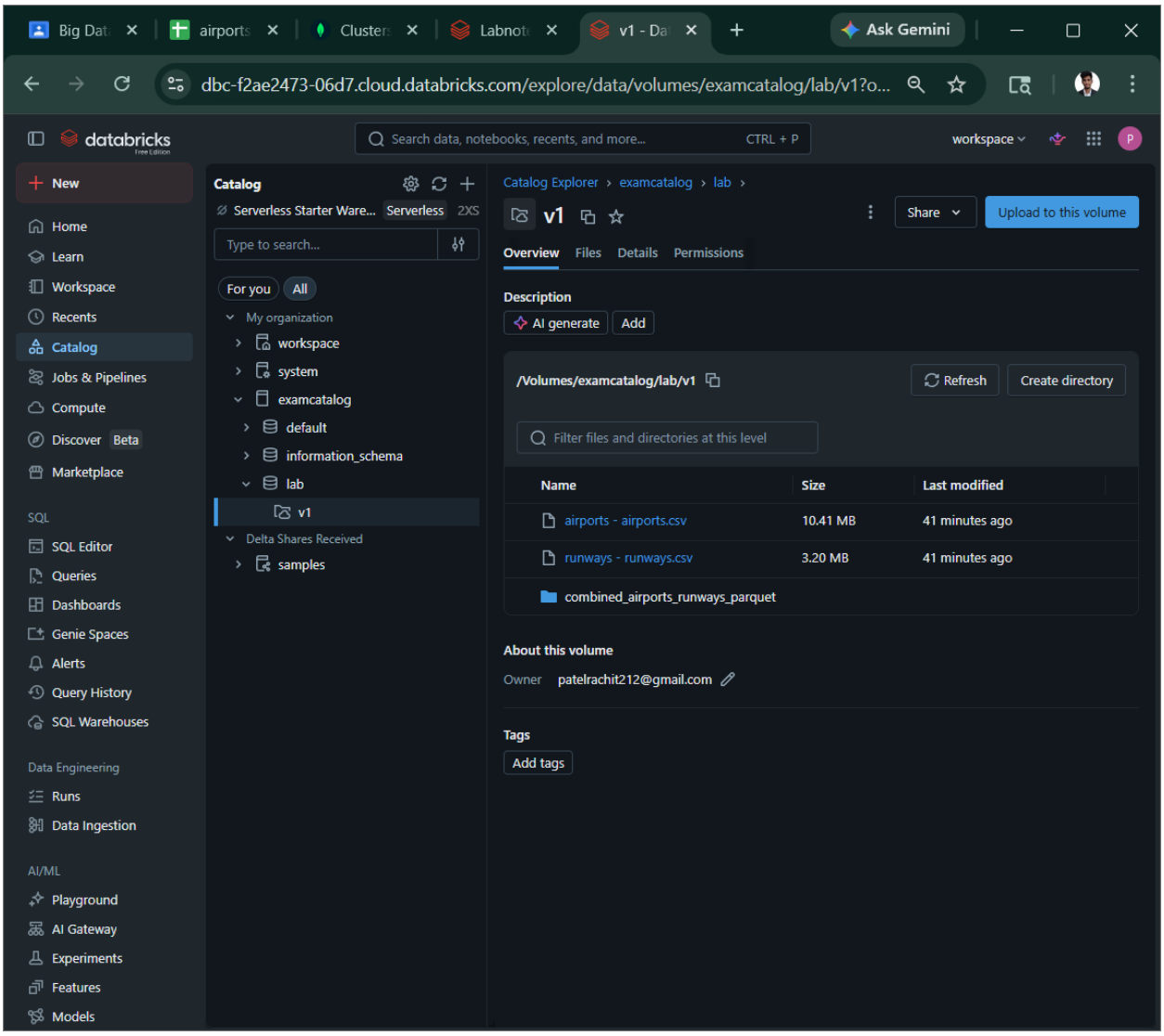
+ Code + Text Genie Code

Just now (1s) 24

```python
df_parquet.show(10)
```

See performance (1)

```
+----------+-----+-------------+--------------------+------------+-------------+------------+---------
+-----------+----------+------------+---------+-----------+-------------+---------+--------+-------+--
-----+
|airport_id|ident|         type|                name|latitude_deg|longitude_deg|elevation_ft|continent
|iso_country|iso_region|municipality|runway_id|airport_ref|airport_ident|length_ft|width_ft|surface|li
ghted|
+----------+-----+-------------+--------------------+------------+-------------+------------+---------
+-----------+----------+------------+---------+-----------+-------------+---------+--------+-------+--
-----+
|      6523|  00A|     heliport|   Total RF Heliport|   40.070985|   -74.933689|          11|       NA
|         US|     US-PA|    Bensalem|   269408|       6523|          00A|        80|       80| ASPH-G|
1|
|      6524| 00AK|small_airport|        Lowell Field|   59.947733|  -151.692524|         450|       NA
|         US|     US-AK|Anchor Point|   255155|       6524|         00AK|      2500|       40|   GRVL|
0|
|      6525| 00AL|small_airport|        Epps Airpark|  34.8647995| -86.77030182|         820|       NA
|         US|     US-AL|     Harvest|   254165|       6525|         00AL|      2100|       90|   TURF|
```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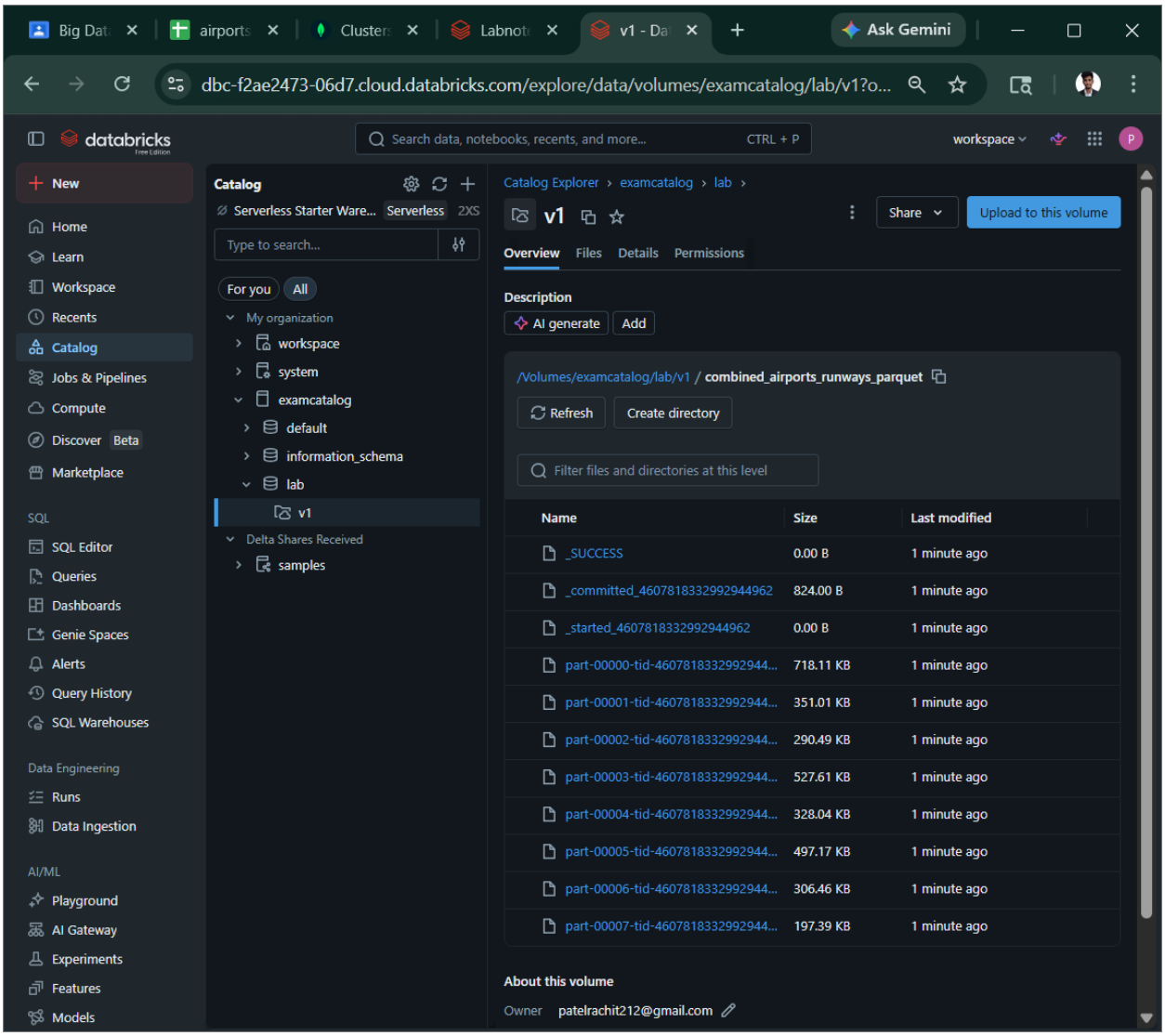
Permissions
Description
AI generate
Add
/Volumes/examcatalog/lab/v1
Refresh
Create directory
Filter files and directories at this level
Name
Size
Last modified
airports - airports.csv
10.41 MB
41 minutes ago
runways - runways.csv
3.20 MB
41 minutes ago
combined_airports_runways_parquet
About this volume
Owner
patelrachit212@gmail.com
Tags
Add tags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Description
AI generate
Add
/Volumes/examcatalog/lab/v1 / combined_airports_runways_parquet
Refresh
Create directory
Filter files and directories at this level
Name
Size
Last modified
_SUCCESS 0.00 B 1 minute ago
_committed_4607818332992944962 824.00 B 1 minute ago
_started_4607818332992944962 0.00 B 1 minute ago
part-00000-tid-4607818332992944... 718.11 KB 1 minute ago
part-00001-tid-4607818332992944... 351.01 KB 1 minute ago
part-00002-tid-4607818332992944... 290.49 KB 1 minute ago
part-00003-tid-4607818332992944... 527.61 KB 1 minute ago
part-00004-tid-4607818332992944... 328.04 KB 1 minute ago
part-00005-tid-4607818332992944... 497.17 KB 1 minute ago
part-00006-tid-4607818332992944... 306.46 KB 1 minute ago
part-00007-tid-4607818332992944... 197.39 KB 1 minute ago
About this volume
Owner patelrachit212@gmail.com